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1.16 Cutting speed, Feed and Depth of cut in lathe

The cutting speed is the distance travelled by a point on the outer surface of the work is one minute.

It is expressed in meters per minute.

Cutting speed = $\frac{\pi DN}{1000}$ m/min

Where,

D – is the diameter of the work in mm

N – is the r.p.m of the work.

Feed

The feed of a cutting tool in a lathe work is the distance the tool advance for each revolution of the work.

Feed is expressed in millimeters per revolution.

Depth of cut

The depth of cut is the perpendicular distance measured from the machined surface to the uncut surface of the workpiece. It is expressed in millimeters.

In a lathe, the depth of cut is expressed as follows

Depth of cut = $\frac{d_1 - d_2}{2}$

Where,

d_1 – diameter of the work surface before machining

d_2 – diameter of the machined surface

1.17 Tools used in a lathe

Tools used in lathe are classified as follows

A. According to the construction, the lathe tools are classified into three types

1. Solid tool
2. Brazed tipped tool
3. Tool bit and tool holders

B. According to the operation to be performed, the cutting tools are classified as

1. Turning tool
2. Thread cutting tool
3. Facing tool
4. Forming tool
5. Parting tool
6. Grooving tool
7. Boring tool
8. Internal thread cutting tool
9. Knurling tool

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